



Quantum Teleportation and Superdense Coding

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Talk Outline

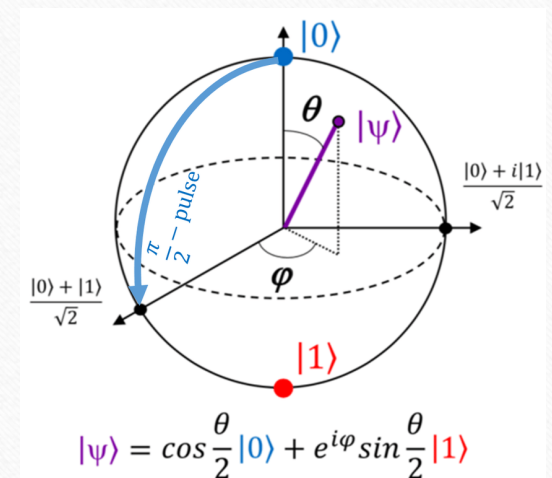
- Motivations and Fundamentals
- Quantum gates and Entanglement generation
- Teleportation Protocol
- Superdense Coding Scheme
- Connections
- Demonstration on Qniverse

Motivation

- Communication is central to both classical and quantum information theory
- Limits of classical channels: bits only, no quantum correlations
- Entanglement acts as a **resource** for communication
- Two protocols that emerge due this resource:
 - Teleportation → sending unknown quantum states
 - Superdense coding → sending classical bits with fewer qubits
- Protocols that arise also lead to important ideas for reducing noise effects and new hardware ideas for building quantum computers.

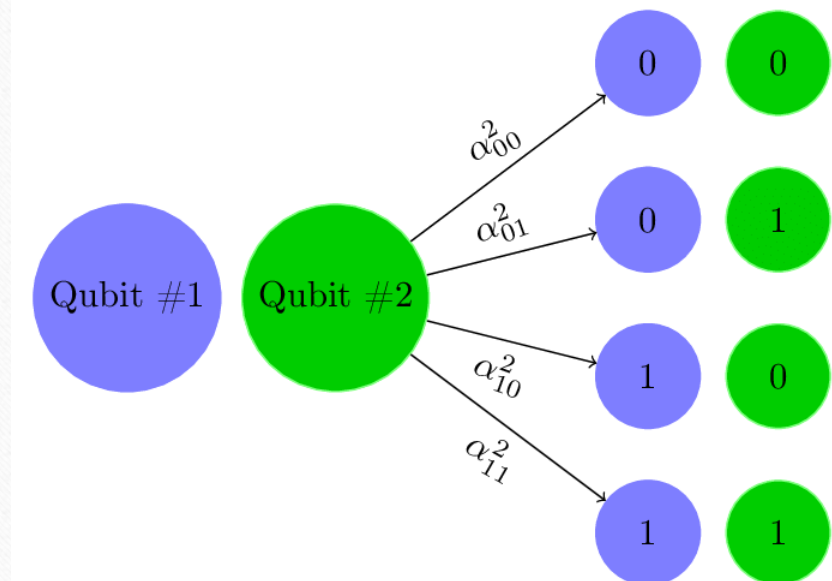
Qubits Refresher

- A qubit = superposition of $|0\rangle$ and $|1\rangle$
- Represented as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- We geometrically represent it on a “Bloch sphere”
- Measurement collapses the state to either $|0\rangle$ or $|1\rangle$



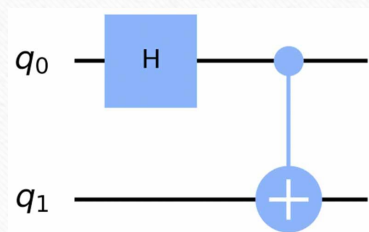
Multi-Qubit Systems

- Combine qubits via tensor product
- Example : $|\psi\rangle \otimes |\phi\rangle =$ two-qubit state
- Basis states: $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$
- Enables correlations beyond classical

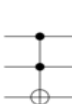


Quantum Gates vs Classical Gates

- Classical: AND, OR, NOT.
- Quantum: Hadamard (superposition), Pauli (X, Y, Z), CNOT (entanglement).
- Quantum circuits = sequence of reversible operations.
- Any computation can be built from a small universal gate set.

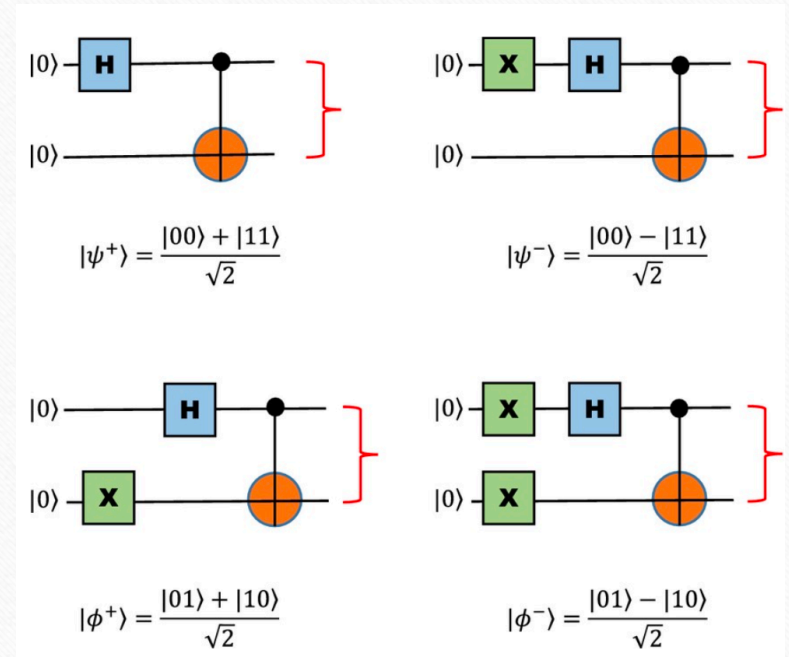


The above Hadamard and CNOT application generates entanglement between q0 and q1

Operator	Gate(s)	Matrix
Pauli-X (X)		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$
Phase (S, P)		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

Entanglement: A unique quantum resource

- Key points: Entangled states (e.g. Bell states: $|\Phi_{\pm}\rangle$, $|\Psi_{\pm}\rangle$) generated by operation : H · CNOT
- Why relevant: Shared Entanglement is the “resource” for several protocols.
- Quick refresh: Entanglement links qubits instantly!
- We will use this link as a ‘quantum channel’



Circuit diagrams to generate all Bell states are given!

Teleportation - Problem

- Suppose Alice wants to send unknown qubit $|\psi\rangle$ to Bob
- But direct measurement destroys the state
- No-cloning theorem: One cannot have a unitary operation to copy a qubit
- Is there a way using entanglement + classical bits?

Classical Limitations

- For the same problem of sending a general state like: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- To ease the restriction: lets assume the sender knows the state completely! $\alpha, \beta \in \mathbb{C}$
- Its imperative to know that to specify the coefficients to arbitrary accuracy, we would require in principle infinite amount of classical information !!
- But a quantum resource based protocol can accomplish this task by sending just two classical bits!
- For a better twist still... the sender even need not know what state she intends to send !

The implications of this as such are pretty clear for secure communication!

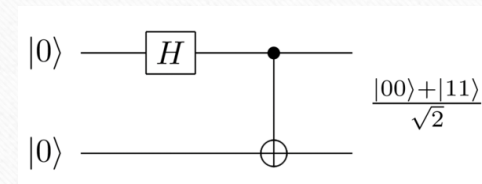
- We will see, how basic quantum mechanics accomplishes this task with ease!
- All we would need is: a few gates, few measurements and ‘some’ classical communication.

Suppose Alice wants to ‘teleport’ $|\psi\rangle$ to Bob.

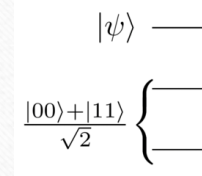
$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

To proceed with teleportation, Alice and Bob additionally require to share an EPR pair between them !

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



So upon sharing the EPR pair they can be depicted as sharing three qubits:

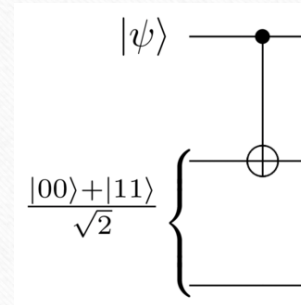
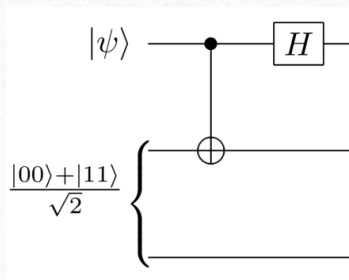


Thus explicitly at the start of the protocol, the initial state is :

$$(\alpha|0\rangle + \beta|1\rangle) \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle}{\sqrt{2}}$$

Suppose now Alice applies CNOT operation on her two qubits, with her qubit of EPR pair as the target state:

Further Alice applies Hadamard gate to her first qubit :



$$\frac{\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle}{\sqrt{2}}$$

$$\frac{\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle}{2}$$

* To analyze the computational basis measurement on first two qubits, we group terms corresponding to each computational basis state for those qubits :

$$\frac{|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)}{2}$$

$$\frac{|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)}{2}$$

Now, when Alice measures her qubits in computational basis, outcome 00 is with probability $\frac{|\alpha|^2}{4} + \frac{|\beta|^2}{4} = \frac{1}{4}$

* And the resultant conditional state for Bob is $\alpha|0\rangle + \beta|1\rangle$, which is indeed what Alice was intending to send to Bob !

We can run through similar calculations on all four outcomes of measurements in Alice's computational basis :

Outcome	Probability	Bob's state
00	$\frac{1}{4}$	$\alpha 0\rangle + \beta 1\rangle = \psi\rangle$
01	$\frac{1}{4}$	$\alpha 1\rangle + \beta 0\rangle = X \psi\rangle$
10	$\frac{1}{4}$	$\alpha 0\rangle - \beta 1\rangle = Z \psi\rangle$
11	$\frac{1}{4}$	$\alpha 1\rangle - \beta 0\rangle = XZ \psi\rangle$

* This indeed is very convenient: As for each case Bob's state is very similar to the original state $|\psi\rangle$ to be teleported !

So to complete the protocol, Alice measures her two qubits in computational basis, with outcomes z and x

Each of z and x are just conventional classical bits, each taking values 0 or 1 .

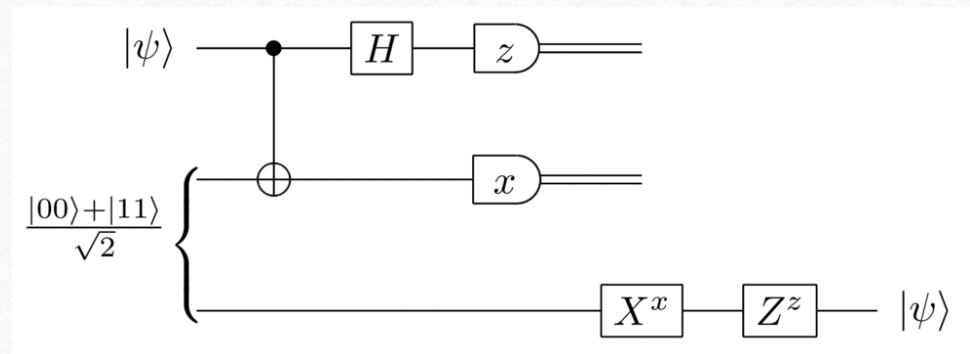
Alice then sends these classical bits (just two !) to Bob.
Who then applies X gate (quantum not gate) if $x = 1$.

$\therefore X^x$ notation !

Similarly Bob applies Pauli-Z gate to his qubit if $z = 1$

$\therefore Z^z$ notation !

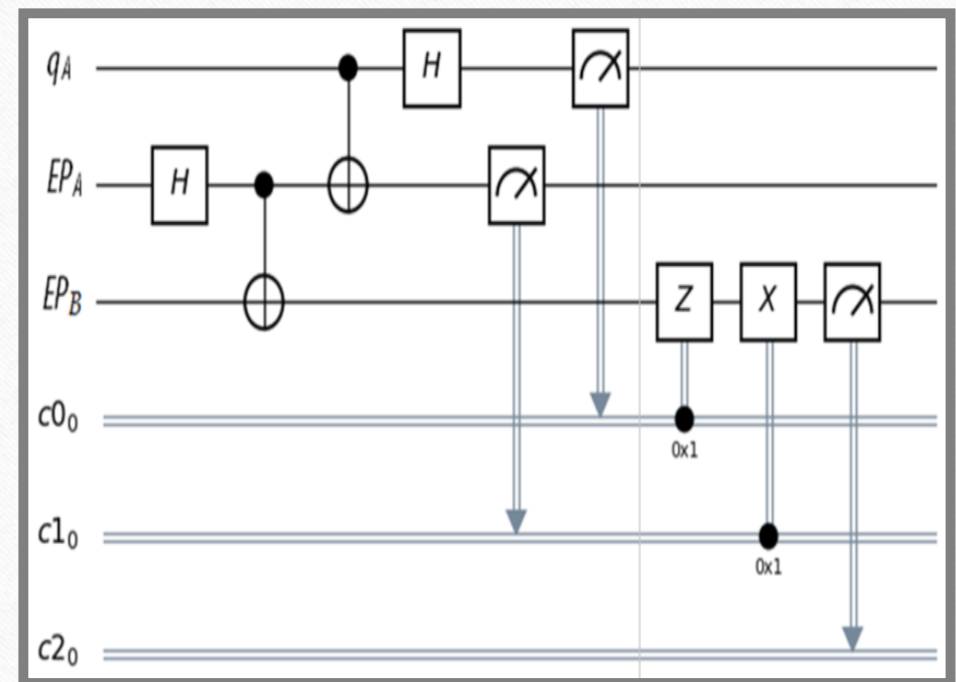
And with such gate implementations, Bob
retrieves the state $|\psi\rangle$, Alice was tasked to send!



And since the state didn't seem to physically travel between Alice and Bob, this is indeed termed as “**Teleportation**”

To Summarize: Teleportation - Circuit

- Alice performs Bell measurement on her two qubits
- 4 possible outcomes (00, 01, 10, 11)
- She sends 2 classical bits to Bob
- Bob applies corrective operation (I, X, Z, XZ)
- Bob's qubit becomes $|\psi\rangle$

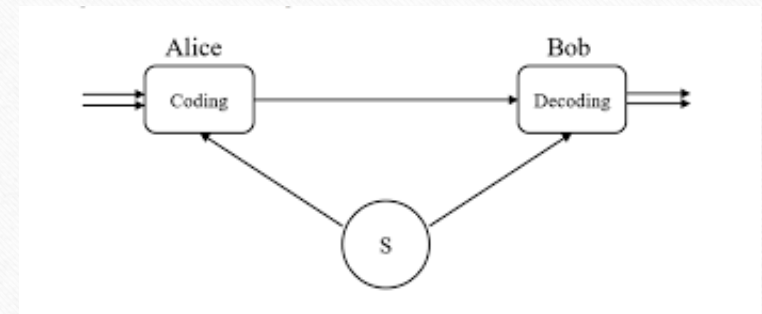


Teleportation – Key Insights

- So the Quantum state got transferred without having to physically travel !
- Classical communication is still necessary though!
- Original qubit got destroyed → consistent with no-cloning
- Widely demonstrated in photonic and ion-trap experiments

Superdense Coding - Problem

- Now: Alice wants to send **two classical bits** to Bob using “one qubit” !?
- Classically : This requires sending 2 bits
- But can she use just one qubit to accomplish this task!?
- Can entanglement act as a resource here and help us in achieving this task!



Superdense: Sending two classical bits using one qubit

- Motivation:
 - In classical communication: 1 bit $\rightarrow$ 1 bit
 - In quantum communication: 1 qubit + Entanglement $\rightarrow$ 2 classical bits
- Connection: Dual of quantum teleportation
 - Teleportation: 1 qubit sent using 2 classical bits
 - Superdense Coding: 2 classical bits sent using 1 qubit

Setup and Encoding

Shared Entanglement:

$$|\phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle) \quad |\psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$$

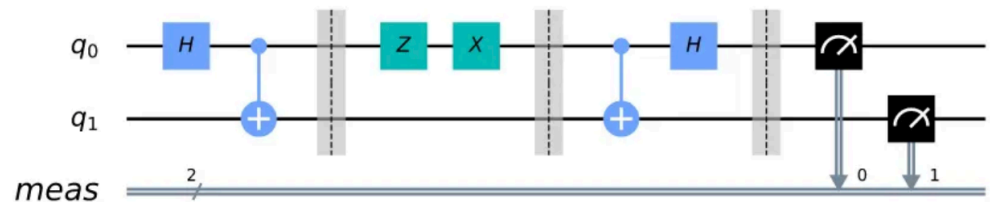
- Alice holds qubit A, Bob holds qubit B

Key Resource: 1 entangled bell pair (ebit)

- Alice intends to send 2 classical bits (possibilities) : 00, 01, 10, 11

- Alice encodes her 2 classical bits by applying a single qubit operator: $U \in \{I, X, Z, XZ\}$

$$\begin{aligned} 00 &\rightarrow I_A |\phi^+\rangle = |\phi^+\rangle \\ 01 &\rightarrow X_A |\phi^+\rangle = |\psi^+\rangle \\ 10 &\rightarrow Z_A |\psi^+\rangle = |\phi^-\rangle \\ 11 &\rightarrow XZ_A |\phi^+\rangle = |\psi^-\rangle \end{aligned}$$



Decoding

- Bob receives Alice's qubit → he now has both qubits in one of 4 bell states
- To decode, Bob performs Bell basis measurement using the circuit :
 - CNOT (Alice's qubit as control, Bob's as target)
 - Hadamard on Alice's qubit
 - Measure both qubits in computational basis

This maps:

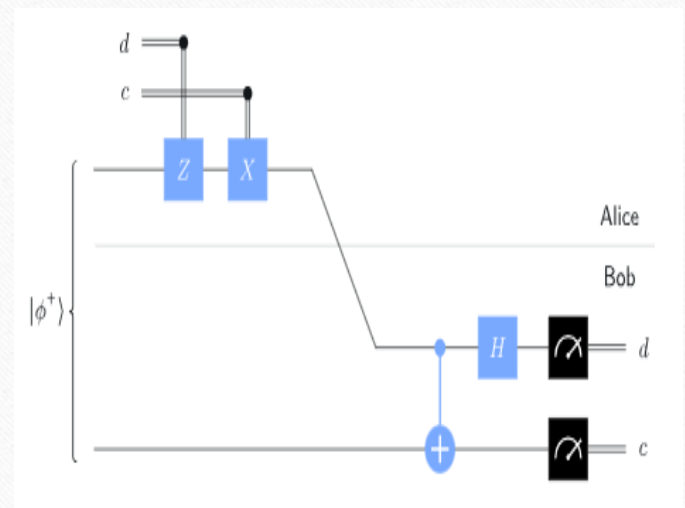
$$|\phi^+\rangle \rightarrow |00\rangle$$

$$|\psi^+\rangle \rightarrow |01\rangle$$

$$|\phi^-\rangle \rightarrow |10\rangle$$

$$|\psi^-\rangle \rightarrow |11\rangle$$

Bob's two classical measurement bits are exactly Alice's original 2-bit message



If $d=1$, Alice performs a Z gate on her qubit A (and if $d=0$ she does not)

If $c=1$, Alice performs an X gate on her qubit A (and if $c=0$ she does not)

Summary and Duality with Teleportation

- Protocol summary:
 - Resource: 1 ebit (pre-shared)
 - Transmission: 1 qubit
 - Payload: 2 classical bits
- Comparison with teleportation:
 - Teleportation: quantum state $\rightarrow$ 2 classical bits required
 - Superdense coding: 2 classical bits $\rightarrow$ 1 quantum bit + entanglement
- Key takeaway: Entanglement doubles the classical communication capacity of a single qubit.

Duality of Protocols

- Teleportation : Quantum $\rightarrow$ (Entanglement + Classical bits)
- Superdense Coding : Classical bits $\rightarrow$ (Entanglement + 1 qubit)
- Both showcase entanglement as a fundamental communication resource

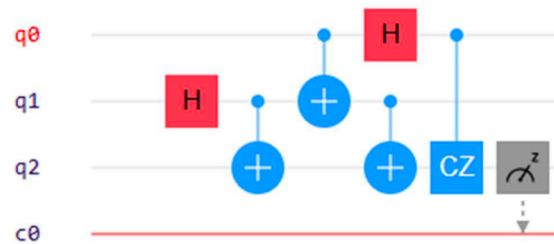
Applications and Extensions

- Foundation for quantum networks and quantum internet
- Quantum repeaters use teleportation
- Superdense coding = efficient information transfer in entanglement-based systems
- Challenges : noise, decoherence, resource management

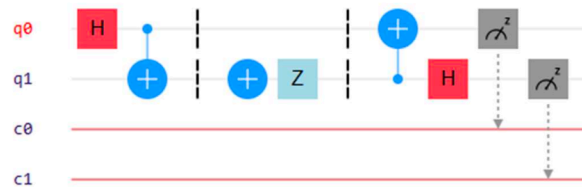
Closing and Outlook

- Quantum communication harnesses entanglement beyond classical limits
- Teleportation and superdense coding → “textbook duals” of quantum info theory
- Experimental progress is leading towards real-world quantum networks
- Future : scalable quantum network

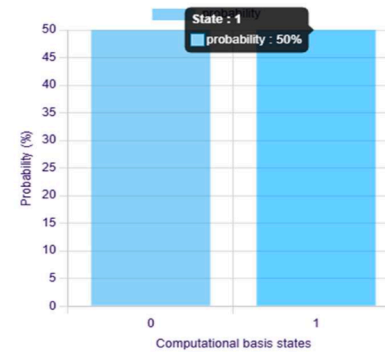
Time to Demonstrate the protocols on Qniverse !



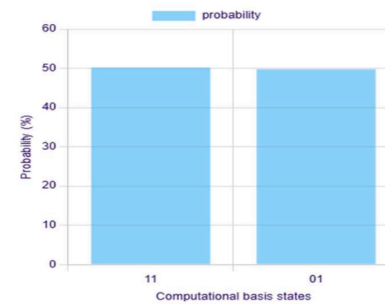
Qniverse circuit implementation for teleporting state : $|\psi\rangle$



Performing superdense coding with Qniverse for any state: $|\psi\rangle$



Even for teleporting state: $|+\rangle$



For superdense coding with state: $|+\rangle$

Thank you for your attention !